CSE400 Lecture 17 Scribe

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Below is a faithful reconstruction of CSE400 Lecture 17: Joint Random Variables an…

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PDF

Generate a lecture scribe intended to serve as exam-reference material for CSE400 Lecture 17: Joint Random Variables and Joint Distributions.

The scribe must faithfully reconstruct the lecture content so that a student can reliably revise the lecture using only this document.

Use only the provided context (lecture slides/PDF and relevant textbook excerpts). Do not introduce any content, explanations, examples, interpretations, or intuition that are not explicitly present in the provided context.

While writing the lecture scribe, reason step-by-step through all definitions, notation, assumptions, derivations, graphical interpretations, and worked examples exactly as presented in the lecture. Ensure that all logical dependencies between ideas are explicitly explained so that the reasoning can be followed during exam revision.

The lecture scribe must clearly and systematically reconstruct the lecture structure, including:

1. Motivation for Joint Random Variables
   - Explanation of why multiple random variables are studied together
   - All real-world examples presented in the lecture (traffic systems, wireless networks, weather, etc.)
   - Interpretation of dependence, independence, correlation, covariance as discussed

2. Discrete Case – Joint PMF
   - Setup of discrete random variables and product space
   - Definition of joint PMF
   - Interpretation of joint probability
   - Joint PMF table representation
   - Properties of joint PMF (non-negativity, summation to 1)
   - Step-by-step reconstruction of the rolling two dice example, including:
     - Definition of variables
     - Construction of joint probability table
     - Observations (such as dependence)

3. Continuous Case – Joint PDF
   - Analogy with discrete case (values, sums vs integrals)

- Definition of joint PDF
- Interpretation using geometric area (small rectangle dx dy)
- Domain of joint random variables (product region)
- Key properties of joint PDF
- Important note that PDF may exceed 1 but represents density

4. Worked Example (Joint PDF)
- Given joint PDF exactly as in lecture
- Step-by-step normalization (validity check using double integrals)
- Visualization of region and event
- Step-by-step computation of required probability
- All intermediate steps exactly as shown

5. Joint Cumulative Distribution Function (Joint CDF)
- Definition using event notation
- Mathematical definition of joint CDF
- Continuous case derivation using double integrals
- Recovery of joint PDF using partial derivatives
- Discrete case formulation using summation
- Interpretation of summation region

6. Properties of the Joint CDF
- Non-decreasing property
- Boundary conditions (lower-left and upper-right limits)
- All properties exactly as presented

Include:
- Definitions and notation exactly as introduced
- Assumptions and conditions
- Step-by-step derivations where present
- Step-by-step worked examples
- Graphical/geometric interpretations where shown
- Logical transitions between concepts

Maintain a formal academic tone suitable for exam preparation.

Do not:
- Add new examples or explanations
- Simplify into tutorial-style content
- Include any material not present in the provided lecture context

Below is a **faithful reconstruction of CSE400 Lecture 17: Joint Random Variables and Joint Distributions**, strictly based on the provided lecture slides L15_L17.
All definitions, derivations, examples, and interpretations follow the lecture exactly, with no added content.

Lecture Scribe: Joint Random Variables and Joint Distributions (Lecture 17)

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1. Motivation for Joint Random Variables

In science and real-life scenarios, we are often interested in **two (or more) random variables simultaneously** L15_L17.

Examples (as presented in lecture):

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members L15_L17

Engineering and System-Based Examples:

Example 1: Smart Transportation

- $X =$ Vehicle Speed
- $Y =$ Traffic Density
- Question: Can speed be high when traffic density is high?
- Conclusion: X and Y are **correlated random variables**
- Applications:
  - Adaptive traffic signals
  - Congestion prediction
  - Autonomous vehicle routing L15_L17

Example 2: Wireless Networks

- $X =$ Signal-to-Noise Ratio (SNR)
- $Y =$ Data Throughput
- Throughput depends on SNR, but also on:
  - Fading
  - Interference
  - Congestion
- Hence, relationship is conditional and not deterministic L15_L17

Example 3: Weather Prediction

- $X =$ Rainfall
- $Y =$ Temperature L15_L17

Example 4: Drone Delivery

- $X =$ Wind Speed
- $Y =$ Delivery Time L15_L17

Example 5: Dice Example

- X and Y from rolling dice
- Events such as $X + Y = 7$, $X > Y$ require joint PMF L15_L17

Key Interpretation

- Joint distributions allow computation of probabilities involving **both variables simultaneously**
- They help understand **relationships between variables**
- If variables are independent → simplest case
- Otherwise → dependence is measured using:
  - **Covariance**
  - **Correlation** L15_L17

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2. Discrete Case – Joint PMF

Setup

Let:

$$X \in \{x_1, x_2, \ldots, x_n\}, \quad Y \in \{y_1, y_2, \ldots, y_m\}$$

The pair (X, Y) takes values in the **product set**:

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

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Definition (Joint PMF)

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

This represents the probability that **both events occur simultaneously** L15_L17.

Joint PMF Table Representation

- Organized as a table with:
  - Rows: values of X
  - Columns: values of Y
- Each entry: $p(x_i, y_j)$

Key Properties

1.

0 \le p(x_i, y_j) \le 1
]
2. $\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$
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Worked Example: Rolling Two Dice

Experiment

- Roll two fair dice

Define

- $X =$ value on first die
- $T =$ total (sum) of both dice L15_L17

Joint Probability Construction

- Each outcome (i, j) has probability:

$$\frac{1}{36}$$

- Construct joint PMF of (X, T)

Table Observations (from page 14):

- Entries are either:
  - 0, or
  - $\frac{1}{36}$
- X and T are **not independent** L15_L17

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3. Continuous Case – Joint PDF

Analogy with Discrete Case

- Discrete values → Continuous intervals
- Joint PMF → Joint PDF
- Summations → Double integrals L15_L17

Domain (Product Region)

If:

$$X \in [a, b], \quad Y \in [c, d]$$

then:

$$(X, Y) \in [a, b] \times [c, d]$$

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Definition (Joint PDF)

A function $f(x, y)$ is a joint PDF if:

$$P((X, Y) \text{ in small rectangle } dx\, dy) \approx f(x, y)\, dx\, dy$$

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Geometric Interpretation

- Probability corresponds to **area of a small rectangle**
- Rectangle dimensions:
  - Width = dx
  - Height = dy L15_L17

Key Properties

1.

f(x,y) \ge 0
]
2. $\int_c^d \int_a^b f(x, y)\, dx\, dy = 1$

Important Note

- $f(x, y)$ **may exceed 1**
- Because it is a **density**, not a probability L15_L17

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4. Worked Example (Joint PDF)

Given

$$f(x, y) = 4xy, \quad 0 \le x \le 1, \quad 0 \le y \le 1$$

- Domain: unit square $[0, 1]^2$ L15_L17

Task 1: Validity (Normalization)

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

Step-by-step:

$$= \int_0^1 \left(\int_0^1 4xy \, dx \right) dy$$

$$= \int_0^1 2y \, dy$$

$$= [y^2]_0^1 = 1$$

Hence:

- $f(x, y) \geq 0$
- Total probability = 1

⇒ **Valid joint PDF** L15_L17

Task 2: Compute Probability

Event:

$$A = \{X < 0.5, \ Y > 0.5\}$$

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy \, dy \, dx$$

Step-by-step:

$$= \int_0^{0.5} \left[2xy^2\right]_{0.5}^1 dx$$

$$= \int_0^{0.5} 2x(1 - 0.25) \, dx$$

$$= \int_0^{0.5} \frac{3}{2} x \, dx$$

$$= \frac{3}{16}$$

Final Answer:

$$P(A) = \frac{3}{16}$$

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5. Joint Cumulative Distribution Function (Joint CDF)

Notation

$$X \leq x, \quad Y \leq y$$

represents:

$$\{X \leq x \text{ and } Y \leq y\}$$

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Definition

$$F(x, y) = P(X \leq x, Y \leq y)$$

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Continuous Case

If joint PDF $f(x, y)$ exists:

$$F(x, y) = \int_c^y \int_a^x f(u, v)\, du\, dv$$

Recovering Joint PDF

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

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Discrete Case

$$F(x, y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j)$$

Interpretation

- Add all probabilities:
  - Rows where $x_i \le x$
  - Columns where $y_j \le y$ L15_L17

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6. Properties of the Joint CDF

1. Non-decreasing

- If x or y increases:

$$F(x, y) \text{ stays same or increases}$$

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2. Lower-left Boundary

$$F(x, y) = 0 \text{ at } (-\infty, -\infty)$$

$$\lim_{(x,y) \to (-\infty, -\infty)} F(x, y) = 0$$

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3. Upper-right Boundary

$$F(x, y) = 1 \text{ at } (\infty, \infty)$$

$$\lim_{(x,y) \to (\infty, \infty)} F(x, y) = 1$$

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End of Lecture 17 Scribe

This document reconstructs the full lecture structure, definitions, derivations, examples, and properties exactly as presented in the provided material.